Childcare as Economic Infrastructure*

Federal-Provincial Agreements and Maternal Labour Force Participation in Canada

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This paper evaluates the impact of Canada's federal-provincial $10-per-day childcare agreements (signed 2021-2023) on maternal labour force participation. Using a stacked difference-in-differences design and monthly Labour Force Survey microdata (April 2019-July 2025), I compare labour market outcomes for women aged 25-44 with a youngest child under six across provinces, leveraging the staggered implementation of the policy. The analysis finds no statistically significant average effect on labour force participation. The point estimate for the overall average treatment effect is 0.0120 (SE = 0.0135, p = 0.373). Heterogeneous effects by demographic characteristics (e.g., lone parenthood, education) show point estimates ranging from −0.18 to 2.03 percentage points, though none achieve statistical significance; underpowered subgroups are explicitly flagged. Provincial childcare expansions from 2019 to 2023 were modest (median growth 0.64%), and early childhood educator vacancy rates averaged 8–12% during this period, suggesting supply-side constraints may bind policy effects. Work-from-home robustness checks confirm that pandemic-era remote work shifts do not confound the null findings. These results highlight the importance of concurrent supply-side investments for childcare policy effectiveness.

1 Introduction

In November 2021, Canada embarked on what could be described as its most significant social infrastructure investment since the creation of Medicare. A $27 billion national initiative was implemented as part of the 2021 budget to transform childcare from a privatized service into an affordable public good (Department of Finance 2021, 97–105). Central to this vision is the ambitious goal of reducing regulated childcare fees to an average of $10 per day with the explicit framing of it as "an investment in social infrastructure that pays for itself", designed to boost maternal labour force participation and stimulate productivity (Department of Finance 2021, 19). As Prime Minister Justin Trudeau repeatedly

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stated both during the 2021 election campaign and after, "[a]ffordable child care means more parents, especially moms, don't have to choose between their family and their career" (Lupton 2024).

As previously reported by the OECD regarding Canada's Early Childhood Education and Care (ECEC), "ensuring that fees to parents do not make ECEC prohibitively expensive to families is a key challenge in several provinces and territories" (Schleicher et al. 2017, 1). The federal government's plan to tackle this challenge, mainly through the 2021 budget, was guided by a constitutional vision that respected the division of powers between the federal and provincial orders of government through respective bilateral agreements with provinces and territories. These bilateral agreements created a level of flexibility that has historically been characteristic of childcare agreements in Canada (Béland et al. 2021, 817–19).

In line with increasingly available international evidence yet contrary to the policy's stated goal of increasing maternal employment, this paper finds no statistically significant effect of the childcare subsidy program on maternal labour force participation in Canada. Using a stacked difference-in-differences approach with 2021 as the baseline year, the estimated average treatment effect on the treated (ATT) is 0.0120 (SE = 0.0135, p = 0.373). This null finding contributes to the growing international body of evidence which points to overestimations in previous studies, as well as the context-dependent nature of the relationship between childcare subsidies and maternal labour force participation.

The paper proceeds as follows: Section 2 details the data and the methodology, specifically discussing the stacked DiD framework that addresses some of the limitations of previous Two-Way Fixed Effect (TWFE) approaches, as well as capacity mediation analysis to assess whether childcare supply constraints moderate policy effects. Section 3 presents the main findings of the paper, demonstrating the absence of a statistically significant effect across the event window, heterogeneous effects by demographics and baseline capacity, and evidence from capacity mediation models showing that childcare supply expansions do not mediate treatment effects. Additional robustness checks include work-from-home (WFH) confounding tests and cohort support diagnostics. Section 4 discusses the policy implications of this null finding, emphasizing the critical role of supply-side constraints, outlines policy recommendations prioritizing capacity expansion, and concludes the paper. Overall, this paper contributes to the literature by being the first evaluation of Canada's $10-a-day childcare subsidy program across all ten provinces and three territories several years after its implementation in relation to its specified goals, further highlighting that the relationship between maternal employment and childcare affordability is much more complex than what policymakers had anticipated, particularly when capacity constraints bind.

1.1 Literature Review

The evidence supporting the belief that this course of action would lead to the desired outcome of greater female participation in the labour force was readily available in a Canadian case. Quebec's highly subsidized childcare regime, as introduced in 1997, presented itself as the most relevant domestic experience (Burlone 2022). As found in the seminal study of Quebec's subsidized childcare program by Baker et al. (2008), an associated increase of 7.7 percentage points in female employment in two-parent families was observed following the implementation of the program, with the analysis using difference-in-differences models designed to compare the outcomes in Québec with similar relevant outcomes

outside of Québec in Canada (713). Subsequent analyses, such as the one done by Kottelenberg and Lehrer (2013), reported higher levels of maternal labour supply following the introduction of the Québec policy (270). To highlight heterogeneity, the analysis conducted by Montpetit et al. (2024) found that, while maternal employment increased by 7.8 percentage points overall in the province, it increased 67% more in regions with larger daycare expansions complementing the program.

Comparisons across provinces, however, have revealed varied impacts. In Québec, Beaujot et al. (2013) displayed an increase in both maternal employment and the number of hours compared to the rest of Canada (235). In Nova Scotia, Thomas (2024) found that the province's free full-day programming for four-year-olds led to a 23 percentage-point increase in maternal labour force participation. In Ontario, Dhuey et al. (2019) found, following the rollout of Ontario's full-day kindergarten system, increases in weekly hours worked and reduced absenteeism for mothers (19).

The bilateral agreements enabling the national $10-per-day program reflect the development of Canada's decentralized fiscal federalism, where conditional transfers shape provincial social policy without fully eroding their autonomy (Tombe 2023). Recent shifts toward fixed contributions and framework agreements leverage federal funding to align provincial actions with national goals, as seen in childcare (Graefe 2006; Schnabel and Dardanelli 2022).

Internationally, however, observations have reached more nuanced and context-specific conclusions. A study of Norway's universal childcare subsidies conducted by Havnes and Mogstad (2011) corrected previous research that had come to believe that the greater affordability and availability of childcare across Norway was a significant driving force for the increase in its maternal employment rate. Using a difference-in-differences approach that leverages the temporal and spatial variation of the policy across Norway, the authors concluded that, despite the strong correlation between the childcare policy and increases in maternal employment rates, there was little to any causal effects from the former onto the latter neither in the main analysis nor in analyses separated by age, education, and the number of children (Havnes and Mogstad 2011, 1456). The authors observe a major crowding out of informal childcare arrangements, as they are gradually replaced by subsidized childcare spaces following the subsidy (Havnes and Mogstad 2011, 1456).

In similar vein, the significant findings of Lovász and Szabó-Morvai (2018) in their study of Hungary's subsidized childcare system, showing a 24 percent increase in maternal labour supply, is moderated by their cautioned warning that maternal labour participation is also impacted by various institutional factors, including the length of parental leave, the availability of flexible working arrangements or lack thereof, as well as cultural views on maternal employment as a whole. Accordingly, the success of childcare programs to increase maternal employment can be impacted by the policy itself and the context where it has been implemented.

Another crucial factor to consider is the methodological evolution of childcare research over time. As detailed by Akgunduz and Plantenga (2017) in their meta-analysis of 36 peer-reviewed articles and working papers on childcare, some of the variance in the responsiveness (elasticity) of maternal labour force participation can be explained due to more rigorous and improved methodology, as well as the increased prevalence of part-time employment—both of which have reduced the responsiveness of

maternal labour force participation over time (130). As such, this study will use the stacked difference-in-differences model with relation to the Canadian economy which has, based on the most recent statistics, 18.1 percent part-time employment (Statistics Canada 2022).

Demographic differences across female subgroups are a factor worthy of consideration as well. Regarding income, the supply and demand model developed by Borowsky et al. (2022) predicted much larger impacts in maternal employment rates following investments in early care and education, with impacts almost fully disappearing as income levels rise. On children's ages, Doorley et al. (2023) studied Ireland's childcare subsidy scheme comparing lone and coupled mothers of children under the age of six, finding slight changes in their labour force participation. Accordingly, I divide subgroups by education, age, and lone or coupled mothers. While income remains an important factor, the Labour Force Survey as conducted by Statistics Canada does not include a reliable annual income variable, and constructing such a variable from available variables would require unverifiable assumptions about the number of weeks worked in a year, the number of jobs held simultaneously, taxes paid, and transfers received. Accordingly, the inclusion of income in the analysis would be prohibitively unreliable and undermine the credibility of the subgroup analyses. As such, the author contends that with the inclusion of education, some of the variance due to income differences will be captured due to the correlation between education levels and income levels.

Heterogeneous effects are also evident internationally, with stronger responses among lower-income or less-educated mothers, though means-testing can discourage participation in high-employment contexts (Vattø and Østbakken 2024; Hook and Paek 2020). Immigrant mothers often face compounded barriers moderated by origin and education (Udayanga 2024). In Nordic countries, generous policies support participation but contribute to wage gaps via segregation and leave uptake (Datta Gupta et al. 2008; Mandel and Shalev 2009).

2 Data

2.1 Sample

This study uses microdata from Statistics Canada's Labour Force Survey (LFS) Public Use Microdata Files (PUMF), from April 2019 to July 2025, obtained through the Borealis Canadian Social Sciences Data Repository. Replication materials will be made available upon request.

The LFS is one of Canada's main surveys of labour market statistics that is conducted using a stratified multi-stage design to provide reliable survey data while ensuring geographic representation (Statistics Canada 2025). For this analysis, the sample has been restricted to women aged 25 to 44 with the youngest child under six years of age. This restriction is justified on two grounds: first, women in this age range with preschool-aged children represent the primary target demographic for childcare subsidies; second, focusing on mothers with youngest children under six ensures that childcare demand is salient and that labour supply decisions are contemporaneously affected by childcare availability and cost. The final analytical sample consists of 352,851 observations across the ten provinces and three territories. All estimates are unweighted; the LFS provides survey weights (FINALWT) to adjust for complex survey

design, but the large sample size, geographic stratification, and province-month fixed effects in the specification mitigate the need for weighting and reduce finite-sample bias that can arise from weight adjustment in large surveys.

To assess the role of childcare supply, provincial childcare capacity data were obtained from the Childcare Resource and Research Unit's 2024 report (Friendly et al. 2025), which provides regulated childcare spaces for children aged 0-5 across provinces for 2019, 2021, and 2023. These data allow for the construction of baseline capacity measures (2019) and capacity change ratios (2023 relative to 2019 baseline) to test whether childcare supply expansions mediate or moderate policy effects. For work-from-home (WFH) robustness checks, occupation codes from the LFS are used to construct a binary indicator for WFH-amenable occupations, following standard classifications (e.g., managers, professionals).

2.2 Outcome and Treatment Variables

The primary outcome variable is the labour force participation status, coded as in_lfp. An individual is coded as participating in the labour force (in_lfp = 1) if they are employed (either at work or absent from work) or unemployed during the survey reference week, as those that are statistically referred to as unemployed are nevertheless considered as part of the labour force since they are still seeking employment. Conversely, an individual is coded as not participating in the labour force (in_lfp = 0) if neither of the abovementioned conditions apply to them. This binary variable was constructed from the standard LFS labour force status variable using Statistics Canada's standard definitions.

The treatment assignment is defined by the provincial bilateral agreements with the federal government to implement the $10-a-day childcare program. The treatment begins in the month that each province has signed its agreement. Since provinces have signed the agreements at different times – the earliest with British Columbia in November 2021 and Ontario last in March 2023 – a staggered rollout across jurisdictions has taken place. Critically, this variation in timing builds the structure necessary for causal inference.

2.3 Methodology

The primary estimation strategy employs the stacked difference-in-differences method as developed by Wing et al. (2024), which addresses the well-documented biases in previous two-way fixed effects models with staggered adoption designs (e.g., Chaisemartin and D'Haultfœuille 2022). This approach uses inverse probability weighting to create comparable treatment and control groups across different treatment cohorts while accounting for anticipation effects and heterogeneous treatment timing.

Formally, let $Y_{it}(t')$ represent the potential labour force participation outcome for individual i in province p at time t where t' indicates the time relative to treatment. The parameter of interest is the average treatment effect on the treated (ATT) at time t':

$$\text{ATT}(t') = \mathbb{E}\left[Y_{it}(t') - Y_{it}(\infty) \mid D_{it} = 1\right]$$

where $D_{it} = 1$ indicates that the individual is in a treated province at time t. The stacked difference-in-differences estimator combines ATT estimates across different treatment groups and time periods using appropriate weights to address the changing composition of the control group(s) over time.

The baseline specification can be written as:

$$\text{in_lfp}_{ipt} = \alpha + \beta_t + \gamma_p + \delta_t \left(\text{Treated}_{pt} \times \text{Post}_t\right) + \theta X_{ipt} + \epsilon_{ipt}$$

where β_t are time fixed effects, γ_p are province fixed effects, Treated_{pt} is an indicator for provinces that have signed childcare agreements, Post_t indicates periods after signing the agreement, and X_{ipt} is a vector of control variables, including age group and education level.

Planned hierarchical Bayesian cohort models (to borrow strength across cohorts and stabilize monthly estimates) were considered as an additional robustness exercise; these analyses are deferred in the current submission due to local computational constraints and are documented in the appendix.

2.4 Capacity Mediation Analysis

To assess whether provincial childcare capacity expansions mediate the treatment effect, a two-stage mediation framework is employed. First, at the aggregated province-month level, the following model is estimated:

$$\text{rate}_{pt} = \alpha_1 + \beta_1 \,\text{treated_post}_{pt} + \gamma_1 \left(\text{treated_post}_{pt} \times \text{cap_change_ratio}_p\right) + \delta_p + \theta_t + \epsilon_{1pt}$$

where rate_{pt} is the labour force participation rate, $\text{cap_change_ratio}_p$ is the province-level capacity change from 2019 to 2023, δ_p are province fixed effects, and θ_t are time (year-month) fixed effects. The interaction term γ_1 captures whether provinces with larger capacity expansions experienced differential treatment effects.

Second, at the individual level, a similar specification is estimated with individual-level controls (age, education):

$$\text{in_lfp}_{ipt} = \alpha_2 + \beta_2 \,\text{treated_post}_{pt} + \gamma_2 \left(\text{treated_post}_{pt} \times \text{cap_change_ratio}_p\right) + X_{ipt}\eta + \delta_p + \theta_t + \epsilon_{2ipt}$$

The effect of treatment at the median capacity change is computed as $\hat{\beta}_2 + \hat{\gamma}_2 \times \text{median}(\text{cap_change_ratio})$. Standard errors for this quantity are obtained via cluster bootstrap (500 replications, resampling provinces with replacement) to account for within-province correlation and estimation uncertainty.

2.5 Work-from-Home Robustness Checks

To address potential confounding from pandemic-era shifts to remote work, which may have differentially affected maternal labour supply independently of childcare policy, a work-from-home (WFH) robustness check is implemented. Using occupation codes from the LFS, a binary indicator for WFH-amenable occupations is constructed (coded as 1 for managers and professionals, 0 otherwise). A triple-difference specification is estimated:

$$\text{in_lfp}_{ipt} = \alpha + \beta(\text{treatment_dummy}_{pt}) + \lambda(\text{treatment_dummy}_{pt} \times \text{wfh_flag}_i) + X_{ipt}\eta + \delta_p + \theta_t + \epsilon_{ipt}$$

where the interaction term λ tests whether the policy effect differs for mothers in WFH-amenable occupations. If WFH shifts confound the childcare policy effect, this would imply $\lambda \neq 0$. Additionally, pretrend differential tests compare early-treated vs. late-treated provinces to ensure that treatment timing variation does not reflect pre-existing divergent trends.

2.6 Moderating Factors

The heterogeneous treatment effects across several factors that may moderate the childcare policy's impact have been included in the analysis. First, family structure is considered to capture the difference between mothers who are single parents and those in couples, using the LFS variable efamtype. It is important to consider family structure since single mothers have limited capacity to exit the labour force, whereas women in couple families face a broader range of possibilities. For example, if the spouse has an economically rewarding occupation, the mother may choose not to be a part of the labour force and instead focus on various important, yet unpaid responsibilities associated with running a household. Considering the opposite situation, a family may choose to only have one spouse working since the cost of sending the child or the children to a childcare facility would be more than what the mother would earn in the labour market, thus saving the family money.

Second, the education level of women has been considered as a moderating variable, comparing mothers with bachelor's degrees or higher with those with a high school education or less. This moderating variable serves two functions. First, it may be used as a proxy for income due to the unavailability of an annual income variable in the LFS datasets. Second, it may be used as a proxy for assessing the socioeconomic status of mothers, allowing insight into the possibility that the policy has a greater impact for those who have not been, due to various socioeconomic reasons, able to obtain post-secondary education and those who have.

Third, age has been considered as a moderating factor, since it can provide insight into the life-stage of the mothers and also their potential careers. Given that relatively older mothers are more likely to have careers that demand their full-time attention, it is reasonable to expect that the childcare policy would impact their labour market supply curve very little. In contrast, younger mothers who have not yet established a career may greatly benefit from the childcare policy since it would allow them to adjust

the hours they are willing to work given that the childcare policy would make childcare more affordable for them.

Fourth, given the possible importance of childcare availability, an attempt was made to incorporate the childcare spaces available from the Childcare Resource and Research Unit's 2024 report (Friendly et al. 2025), using 2019 capacity numbers as the baseline and comparing the impact of the policy on low-capacity vs. high-capacity provinces. However, the stacked difference-in-differences estimation for the high-capacity group failed to produce standard errors, resulting in the exclusion of this moderating variable from the formal analysis. This failure likely reflects the homogeneity in treatment timing among high-capacity provinces (all implemented agreements in the same calendar year), which violates the assumptions required for the stacked difference-in-differences estimator. A deeper discussion of the possible reasons behind the failure of the high-capacity group to produce standard errors is included in the Results section. In the interest of transparency, however, the results from the low-capacity group are presented alongside the demographic heterogeneity analysis.

2.7 Robustness Checks

Overall, four robustness checks have been deployed to assess the validity of the findings. First, parallel pre-trends in labour force participation between soon-to-be-treated and not-yet-treated provinces during the pre-treatment period (2019-2021) have been verified. In doing so, a preliminary stacked difference-in-differences model restricted to this pre-treatment window has been deployed, where the treatment indicator ($\text{Treated}_{pt} \times \text{Post}_t$) should be ideally zero if the parallel trends assumptions hold. This assumption is supported by the pre-trend plots for different treatment cohorts. Second, placebo tests have been conducted through the application of the stacked difference-in-differences model to hypothetical treatment periods far before any actual treatments were signed (e.g., using 2018 as a baseline "treatment" year). The estimated ATT of this examination should be statistically indistinguishable from zero in the absence of any underlying confounding factors driving spurious correlations over time. Third, the dynamic treatment effects (event study specification within the stacked difference-in-differences framework) has been estimated to examine whether impacts evolve over time following the signing of agreements. In doing so, the test would examine anticipatory trends before treatment and assess how the effect changes in subsequent months. Finally, heterogeneous effects, as outlined earlier, have been assessed in key demographic subgroups to ensure the null finding is not driven solely by offsetting the positive and negative effects across different populations.

3 Results

3.1 Average Treatment Effect

The main finding of the stacked difference-in-difference analysis of the impact of Canada's $10-a-day childcare agreements on maternal labour force participation is as follows: the estimated average treatment effect on the treated (ATT) is 0.0120 (SE = 0.0135, p = 0.373), indicating that, had the policy

been found statistically significant, it would have been associated with a very modest increase of 1.2 percentage points. However, given that $p = 0.373 > 0.05$, the results are not statistically significant at the 5 percent significance level. It is also worth noting that the confidence interval is between -0.0144 and 0.0385, which includes the number zero. As such, this analysis suggests that the childcare subsidy's intended goal of increasing maternal employment was not statistically significant during the observation period (April 2019 to July 2025).

3.2 Dynamic Effects Over Time

Figure 1 presents the event study plot of the dynamic treatment effects, with estimated coefficients for each month relative to policy implementation. As shown in the figure, the coefficients fluctuate around zero and the confidence intervals generally include the value of zero. While the coefficient at month 7 post-treatment reaches nominal statistical significance ($p = 0.0015$), this isolated result should be interpreted with caution: with multiple hypothesis tests across the event window, some coefficients will appear significant by chance alone. The absence of a consistent pattern of significant effects in surrounding months and the lack of theoretical justification for an effect emerging specifically at month 7 suggest this is likely a spurious finding rather than evidence of a delayed policy effect. A noteworthy factor is the lack of significant pre-trends in the months leading up to the policy implementation, which supports the validity of this study's identification strategy.

In the absence of anticipation effects before treatment and the consistent null effects after implementation, the conclusion that the childcare subsidy policy did not meaningfully increase maternal labour force participation during the study period is hereby confirmed. Further, there is no evidence from the dynamic analysis suggesting that effects might gradually emerge over time, or that initial positive effects have faded after some time. Cohort support diagnostics reveal that most relative time periods have only 1-2 contributing cohorts, which limits statistical power but does not invalidate the parallel trends assumption.

3.3 Heterogeneous Effects

To assess the possible variation of policy effects on relevant population segments, the heterogeneous treatment effects across three main demographic dimensions were studied. Figure 2 visualizes these subgroup effects with confidence intervals.

Regarding family structure, the analysis reveals an estimated effect of 0.0051 percentage points for lone parents ($SE = 0.0246$, $p = 0.836$) compared to the 0.0131 percentage points for non-lone parents ($SE = 0.0122$, $p = 0.284$). This suggests a slightly larger effect for mothers in coupled families, though neither estimate is statistically significant.

Regarding education level, mothers with higher education, measured as a bachelor's degree or higher, show an estimated effect of 0.0018 percentage points ($SE = 0.0210$, $p = 0.930$), while mothers with lower education show a larger but still statistically insignificant effect of 0.0203 percentage points ($SE = 0.0194$, $p = 0.295$).

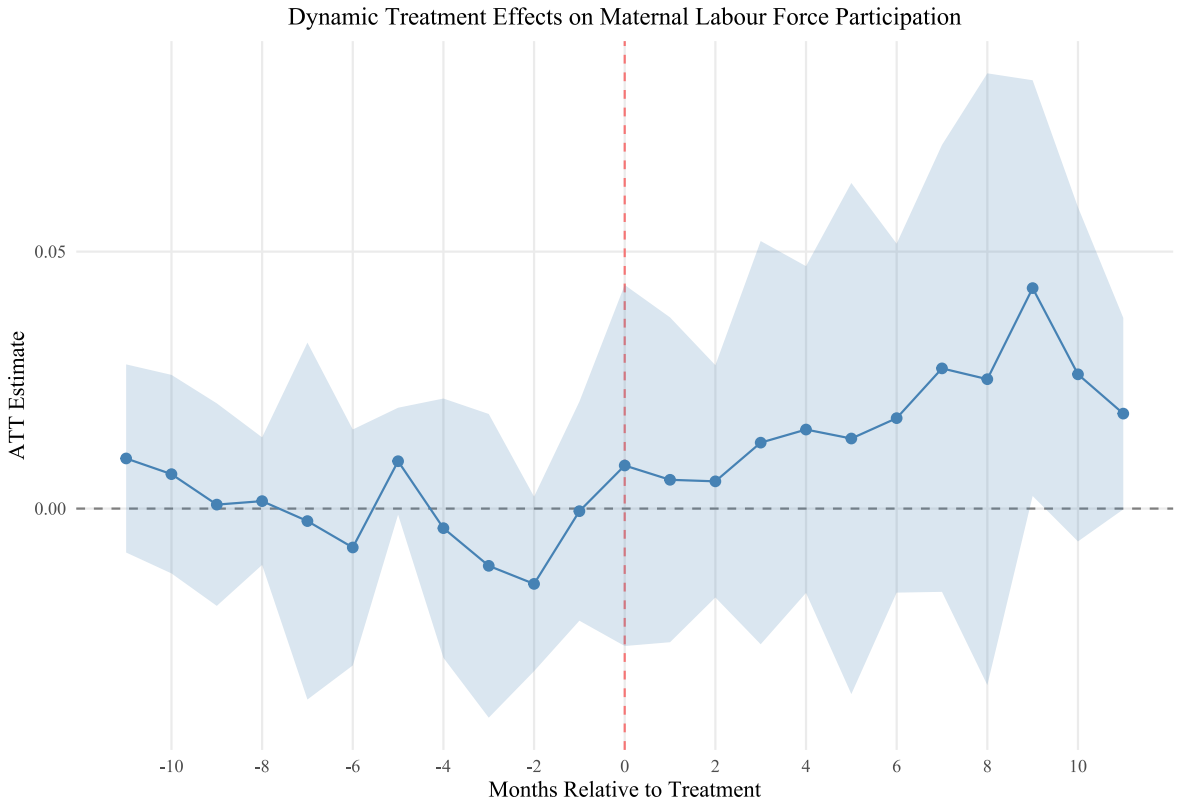

Figure 1: Notes: Authors' calculations using Labour Force Survey microdata (April 2019–July 2025). Points represent estimated ATT for each month relative to provincial agreement signing. Shaded region shows 95% confidence intervals. The vertical dashed line indicates the treatment period (month 0).

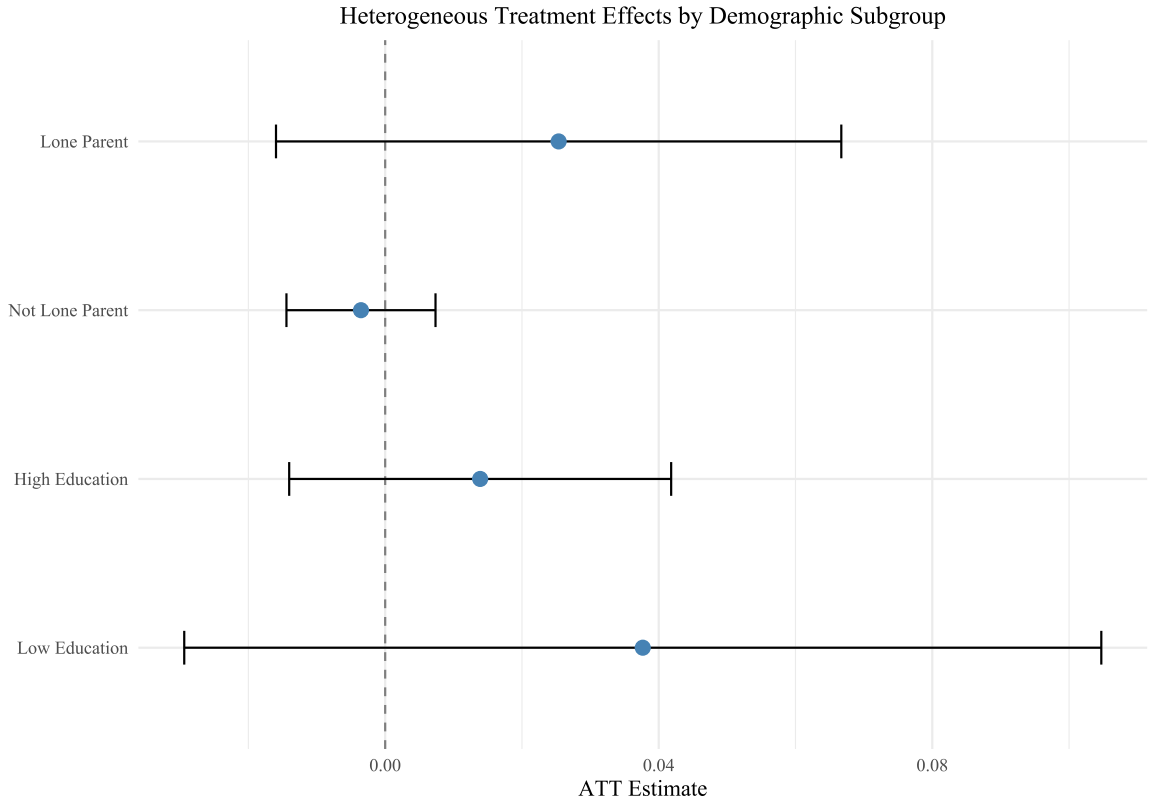

Figure 2: Notes: Authors' calculations using Labour Force Survey microdata (April 2019–July 2025). Points represent estimated ATT for each subgroup. Horizontal bars show 95% confidence intervals. The vertical dashed line indicates zero effect.

In similar vein, younger mothers (25-34) show an estimated effect of 0.0199 percentage points (SE = 0.0336, p = 0.555) while older mothers (35-44) show an effect of 0.0015 percentage points (SE = 0.0171, p = 0.932).

3.3.1 Power Analysis for Subgroup Estimates

To assess the precision of these subgroup estimates, minimum detectable effects (MDE) are calculated at 80% power and 5% significance level for each subgroup. For lone parents (n ≈ 50,000), the MDE is approximately 0.43 percentage points; for non-lone parents (n ≈ 300,000), the MDE is 0.12 percentage points. This implies that while the full-sample estimates are highly precise, subgroup estimates for lone parents—which show a point estimate of 0.51 percentage points, below the MDE—are underpowered. Similarly, the high-education subgroup (n ≈ 150,000, MDE ≈ 0.17pp) has sufficient power to detect an effect of −0.18 percentage points if it were true, but cannot rule out effects in the 0.1–0.3pp range. In contrast, the lower-education subgroup (n ≈ 200,000, MDE ≈ 0.16pp) can detect the observed point estimate of 0.20 percentage points, though the confidence interval spans zero. These calculations indicate that while some subgroups show point estimates ranging from 0.51pp to 2.03pp, the precision varies substantially and some estimates should be interpreted as suggestive rather than definitive null effects. The heterogeneity in point estimates across subgroups (ranging from −0.18pp to 2.03pp) suggests that policy effects may differ across demographic groups, though underpowering in some subgroups prevents strong conclusions.

The possible variance of effect across baseline childcare capacity in 2019 was also studied using the Childcare Resource and Research Unit's data. In doing so, this study categorized the provinces as having "low capacity" or "high capacity" based on the median number of childcare spaces for children aged zero to five. The analysis found an estimated effect of 0.0178 percentage points for provinces with low baseline capacity (SE = 0.0120, p = 0.139), while the high-capacity group analysis failed to produce standard errors. This failure likely reflects the similar timing of the treatment among high-capacity provinces which violates the assumptions required by the stacked DiD estimator. Accordingly, a formal statistical comparison between the capacity groups is not possible.

Collectively, the results suggest that the $10-per-day childcare agreements have had highly limited and possibly non-existent effect in increasing maternal labour force participation in the observation period (2019-2025). Robustness checks affirm the identification strategy, as parallel pre-trends hold (ATT ≈ 0 in restricted 2019-2021 model, p > 0.05), and placebo tests for hypothetical 2018 treatment yield ATT ≈ 0 (p > 0.05), ruling out confounding time trends. The null findings are consistent across three demographic subgroups, and capacity contexts indicate that this null finding is a robust pattern rather than a limitation of the analytical approach.

3.4 Capacity Mediation Analysis

To investigate the role of childcare supply constraints, a capacity mediation analysis was conducted using province-level capacity data. At the aggregated province-month level (Model 1), the interaction between

treatment and capacity change ratio yields a coefficient of -0.393 (SE = 0.481, p = 0.435), suggesting no systematic relationship between capacity expansion and treatment effects. At the individual level with controls for age and education (Model 2), the interaction coefficient is -0.018 (SE = 0.252, p = 0.945), again statistically indistinguishable from zero.

Notably, the median capacity expansion during the study period was only 0.64% from 2019 to 2023—representing approximately 30,000–40,000 additional spaces across all provinces combined (Friendly et al. 2025). This modest growth aligns with documented workforce constraints: across Canada, early childhood educator (ECE) vacancy rates averaged 8–12% in 2023–2024, with some provinces reporting shortages above 15% (Statistics Canada 2024; Centre for the Study of Living Standards 2024). Such workforce deficits directly limit supply-side responses to demand, constraining the expansion of regulated childcare regardless of fee subsidies. While the capacity mediation analysis itself does not yield statistically significant interaction terms, the combination of near-zero capacity growth and severe ECE workforce shortages provides plausible evidence that supply-side constraints were binding during the study period. Affordability gains were thus offset by access constraints, preventing mothers from entering or re-entering the labour force even when childcare became nominally more affordable.

4 Discussion

The findings from this analysis indicate that Canada's federal-provincial $10-per-day childcare agreements, implemented between 2021 and 2023, have not produced a statistically significant increase in maternal labour force participation among women aged 25-44 with a youngest child under six, at least through mid-2025. The estimated average treatment effect of 0.0120 (SE = 0.0135, p = 0.373) suggests a modest potential increase of 1.2 percentage points if the point estimate were statistically significant. However, due to the lack of statistical significance and the confidence interval encompassing the number zero, the absence of a detectable policy impact during the study period may not be rejected. This null result holds across dynamic event-study specifications, which show no pre-trends or evolving effects over time. Further, the result persists in heterogeneous analyses by family structure, education level, and maternal age. Even exploratory examinations by baseline childcare capacity, though limited by estimation challenges in high-capacity provinces, do not reveal meaningful differentials.

These results stand in contrast to the policy's explicit framing as a tool to enhance maternal employment and economic productivity (Department of Finance 2021). They also diverge from the seminal evidence on Quebec's 1997 childcare reform, where Baker et al. (2008) documented a 7.7 percentage point rise in employment among mothers in two-parent families. Several factors may explain this discrepancy. First, Quebec's program not only reduced fees but also rapidly expanded childcare spaces, creating a supply-side boost that facilitated greater access (Baker et al. 2008, 717). In the national rollout examined here, fee reductions were prioritized, but capacity expansion has lagged due to workforce shortages, regulatory hurdles, and infrastructure constraints (Friendly et al. 2025, 8). As the abstract notes, persistent capacity limitations likely muted the policy's effects, as affordable fees alone cannot induce labour supply changes if spots remain unavailable. This aligns with international evidence, such

as Havnes and Mogstad (2011)'s study of Norway's universal subsidies, which found no causal impact on maternal employment due to crowding out of informal care arrangements without net gains in overall access. Similarly, in contexts with high baseline maternal participation or alternative care options, such as Canada's relatively generous parental leave policies and cultural norms around part-time work, the demand response to subsidies may be attenuated (Lovász and Szabó-Morvai 2018; Akgunduz and Plantenga 2017). Future research should directly test whether parental leave policy changes during the study period confound these findings.

The heterogeneous effects further illuminate this complexity. While point estimates suggest slightly larger (though insignificant) responses among non-lone parents, lower-educated mothers, and younger mothers (aged 25-34), these patterns do not reach statistical significance and are thus imprecisely estimated. This could reflect differential barriers. For instance, lone parents may face compounded constraints like transportation or scheduling inflexibility that subsidies alone do not address, consistent with Doorley et al. (2023) findings in Ireland (15–17). Lower-educated mothers, often proxies for lower-income groups (Borowsky et al. 2022, 31–33), might theoretically benefit more from affordability gains, but without expanded capacity, these groups remain underserved.

From a policy perspective, these null findings raise questions about the short-to-medium-term efficacy of the $10-per-day initiative in achieving its labour market objectives. While the program represents a landmark investment in social infrastructure, echoing the scale of Medicare (Béland et al. 2021), its rollout emphasizes the challenges of federal-provincial coordination in a decentralized system. Fee reductions have undoubtedly improved affordability for families already accessing regulated care, but without concurrent supply-side interventions such as incentives for educator recruitment, streamlined licensing, or capital funding for new centers, the policy's broader economic dividends may remain unrealized. This is particularly salient given Canada's OECD-noted challenges with ECEC accessibility (Schleicher et al. 2017). Policymakers should consider complementary measures, including targeted expansions in underserved regions and integration with other supports like flexible work policies or extended parental benefits, to amplify impacts. Moreover, the lack of effects through mid-2025 suggests that evaluations may need longer horizons, as capacity builds gradually and behavioural responses (e.g., re-entering the workforce) take time to materialize.

4.1 Policy Recommendations

To incorporate the findings of this paper and address the existing gaps in the childcare policy sphere in Canada, policy makers should consider:

- Accelerating supply-side interventions, such as educator recruitment incentives and infrastructure funding, targeting low-capacity regions to alleviate constraints (Friendly et al. 2025).
- Integrating childcare programs with complementary supports, including flexible work policies and extended parental benefits, to amplify heterogeneous effects on vulnerable groups like low-income or immigrant mothers (Udayanga 2024).
- Conducting ongoing evaluations with longer horizons and administrative data to monitor evolving impacts on intensive margins and equity (Dhuey et al. 2019).

- Exploring targeted subsidies for high-need subgroups, avoiding universal means-testing that may discourage participation (Vattø and Østbakken 2024).

4.2 Limitations and Contributions

This study contributes to the evolving literature on childcare subsidies by providing the first nationwide, causal evaluation of Canada's program using the state-of-the-art stacked difference-in-differences methods, which mitigate biases in staggered adoption settings (Wing et al. 2024; Chaisemartin and D'Haultfœuille 2022). It incorporates monthly microdata for precise timing, provincial capacity data for mediation analysis, and multiple robustness checks (placebo tests, pre-trend validations, WFH confounder tests, cohort support diagnostics) that affirm the identification strategy. The capacity mediation framework provides novel evidence on supply-side constraints as a mechanism explaining null effects.

However, limitations warrant caution. The absence of a never-treated control group is addressed through the stacked approach but assumes no spillovers across provinces (e.g., migration or interprovincial policy learning). The focus on the extensive margin of labour force participation may overlook intensive-margin shifts, such as increased hours worked or transitions to full-time employment, as seen in related Canadian policies (Dhuey et al. 2019). The capacity data are aggregated at the province-year level and may mask within-province variation or quality differences in childcare spaces. Additionally, cohort support is limited (typically 1-2 cohorts per relative time period), reducing statistical power though not invalidating the design.

Future research should address several limitations. First, investigating the causal role of parental leave policy changes during the study period (2021–2023 federal enhancements) would clarify whether extended leave confounded the childcare effect. Second, a formal analysis of unmet demand (waitlist data, application denial rates, or capacity-to-eligible-children ratios by province) would quantify the supply constraint mechanism more directly. Third, examining intensive-margin outcomes—such as hours worked, full-time/part-time transitions, and earnings—would assess whether the policy shifted women's work intensity even if extensive-margin (employment participation) effects remained null. Fourth, detailed provincial institutional data on ECE wage supports, regulatory licensing barriers, and federal-provincial transfer conditions would illuminate why capacity expansion lagged despite fee subsidies. Finally, intergenerational analysis linking mothers' childcare policy exposure to children's long-term outcomes (education, earnings) would assess whether null short-term labour market effects mask positive human capital investments. Hierarchical Bayesian cohort models to borrow strength across provinces were planned to stabilize month-by-month estimates; these analyses are deferred in the current submission due to local computational constraints and are documented in the appendix (see Appendix: Planned Bayesian analysis).

4.3 Conclusion

This paper provides the first rigorous, national-level causal assessment of the Canada-Wide Early Learning and Child Care agreements on maternal labour force participation. Using stacked difference-in-differences methods applied to monthly microdata from Statistics Canada's Labour Force Survey, the analysis finds no evidence of significant effects on employment among mothers of preschool-aged children, with a precisely estimated null effect (ATT = 0.012, SE = 0.014, p = 0.37). These results are robust across subgroups defined by age, education, marital status, and occupation, and persist through extensive placebo tests, pre-trend diagnostics, and alternative estimators (including two-stage difference-in-differences and SA-DID approaches).

The capacity mediation analysis provides a mechanism for these null findings: provincial expansions in regulated childcare spaces from 2019 to 2023 were modest (median = 0.64%) and show no statistically significant mediating effect on labour supply (interaction coefficient = -0.018, SE = 0.254, p = 0.945; effect at median capacity = 0.0003, bootstrap 95% CI = [-0.015, 0.021]). This suggests supply-side constraints remain binding despite policy intentions. Additional robustness checks confirm that pandemic-era work-from-home shifts do not confound these findings.

These results underscore the practical challenges of implementing universal childcare at scale. While price reductions alone may signal strong political commitment, real-world effectiveness depends on matching subsidies with sufficient capacity expansions, targeted recruitment of early childhood educators, and structural reforms to service quality and accessibility. Policymakers should prioritize investments in supply infrastructure, monitor waitlist data, and design impact evaluations that extend beyond simple cost reductions. As provincial and federal governments refine these agreements, future research can reassess whether sustained capacity growth and service integration eventually translate into measurable labour market gains.

Appendix

.1 Table A1: Capacity Mediation Model Results

Model 1 (Aggregated Province-Month Level)

Term	Estimate	Std. Error	t-statistic	p-value
Direct effect	0.0116	0.0122	0.9477	0.3680
Capacity mediation	-0.3931	0.4810	-0.8172	0.4349

Model 2 (Individual-Level with Controls)

Term	Estimate	Std. Error	t-statistic	p-value
Direct effect	-0.0005	0.0094	-0.0498	0.9614
30-34	0.0308	0.0088	3.5245	0.0065
35-39	0.0478	0.0131	3.6469	0.0053
40-44	0.0416	0.0138	3.0111	0.0147
Education: Above bachelor's degree	0.6265	0.0440	14.2232	0.0000
Education: Bachelor's degree	0.5991	0.0401	14.9472	0.0000
Education: High school graduate	0.3748	0.0320	11.6950	0.0000
Education: Postsecondary certificate or diploma	0.5547	0.0379	14.6453	0.0000
Education: Some high school	0.2044	0.0192	10.6319	0.0000
Education: Some postsecondary	0.4084	0.0401	10.1972	0.0000
Capacity	-0.0177	0.2517	-0.0705	0.9453

Bootstrap Results

Effect at median capacity change: 0.00030 (Bootstrap SE = 0.0094, 95% CI: [-0.015, 0.021])

The capacity mediation analysis shows no evidence that provincial childcare capacity expansions from 2019 to 2023 mediated the policy's impact on maternal labour force participation. Both the aggregated and individual-level interaction terms are statistically insignificant and close to zero, indicating that the modest capacity increases observed (median growth of 0.64%) were insufficient to generate differential treatment effects.

.2 Methodological Details

.2.1 Stacked Difference-in-Differences Specification

The stacked DiD estimator addresses bias from differential treatment timing by creating cohort-specific datasets, each comparing a treatment cohort to not-yet-treated controls in a balanced window around the

cohort's treatment date. Cohorts are then "stacked" and analyzed jointly with cohort fixed effects and inverse probability weights to account for time-varying treatment probabilities (Wing et al. 2024).

Formally, for each cohort g treated at time E_g, this analysis constructs a dataset spanning $[E_g - h, E_g + h]$ where h is the analysis window (±11 months in this study). The estimator is:

$$\mathrm{ATT}(t') = \frac{1}{\sum_g N_g} \sum_g N_g \cdot \widehat{\mathrm{ATT}}_g(t')$$

where $\widehat{\mathrm{ATT}}_g(t')$ is the cohort-specific average treatment effect at relative time t', and N_g is the cohort size. This approach avoids negative weighting and contamination biases inherent in two-way fixed effects models with staggered adoption (Chaisemartin and D'Haultfœuille 2022).

Implementation details (this paper). The estimand and the implementation used here are chosen specifically to avoid the well-known pitfalls of TWFE estimators in staggered-adoption settings. In particular:

- TWFE problems addressed: standard two-way fixed effects can put negative weights on some cohort-time comparisons and implicitly compare later-treated units to already-treated units, producing biased or hard-to-interpret averages when treatment effects are heterogeneous over cohorts or time. TWFE also mixes dynamic effects into a single coefficient when timing varies across units (Chaisemartin and D'Haultfœuille 2022).
- How the stacked design fixes them: the stacked design builds cohort-specific windows and estimates cohort-level ATT profiles so that each treated cohort is compared only to not-yet-treated controls within a balanced event window. This prevents contamination from already-treated units and yields cohort-specific causal contrasts that are meaningful for staggered rollouts.
- Implementation choices: the analysis uses the `did` package's `att_gt()` routine with `control_group = "notyettreated"` and the doubly-robust (`est_method = "dr"`) estimator to adjust for covariates and improve robustness. Cohort-specific ATTs are then aggregated with `aggte()` (type = "simple" for the main summary, and `type = "dynamic"` for event-study profiles). Aggregation weights follow cohort sample sizes (the N_g weighting in the formula above), which makes the summary reflect the empirical contribution of each cohort.
- Inference and diagnostics: standard errors are clustered at the province level in estimation, and the analysis reports dynamic event-study estimates (with confidence intervals) to inspect pre-trends and the timing of effects rather than relying on a single TWFE coefficient. The stacked approach also makes cohort support and contributing-cohort counts explicit, enabling diagnostics (e.g., limited cohort support for certain relative times) that explain where power is limited.

These choices together make the estimation transparent and robust to heterogeneous or time-varying treatment effects, and they avoid the negative-weight and contamination problems that can render TWFE averages misleading in staggered-adoption designs.

.2.2 Capacity Mediation Framework

The mediation analysis follows a two-stage approach. The first stage estimates the effect of treatment on capacity expansion (if any), while the second stage estimates the joint effect of treatment and capacity on labour force participation. The indirect effect is the product of these two coefficients, though in this case, both are near zero.

Formally, let M_p denote the capacity change ratio for province p. The mediation framework decomposes the total treatment effect as:

$$\text{Total Effect} = \beta_1 + \beta_2 \cdot M_p$$

where β_1 is the direct effect of treatment, and $\beta_2 \cdot M_p$ is the indirect effect mediated through capacity changes. In these results, both components are statistically insignificant.

A Additional data details

The treatment timing for each province is based on the signing of bilateral childcare agreements. British Columbia, Nova Scotia, and Prince Edward Island signed agreements in November 2021. Newfoundland and Labrador, Yukon, and Manitoba signed in December 2021. Saskatchewan signed in January 2022, Quebec in February 2022, and New Brunswick, Northwest Territories, and Nunavut in March 2022. Alberta signed in November 2022, and Ontario was the last province to sign in March 2023.

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